

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 1 of 10

(1) Research Program/Project/Study Title: Microbial Quality, Water Activity, and NaCl as Salt Content of Traditionally and Mechanically Processed Dried Fish Products	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Agri-Processing/Fish	(8) Discipline/Field of Specialization Food Process/Agricultural Engineering

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 2 of 10

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

Dried fish is a rich source of protein, lipid and minerals that can serve as a promising source of nutrients to alleviate malnutrition in low-income countries. However, since the drying is often performed with poor hygienic conditions, the microbial quality of such products is uncertain. Mechanical drying and process standardization may improve the microbial quality of the dried fish. Hence, this study aimed to assess the water activity, salt content and microbial load of locally prepared dried fish using traditional brining and sun-drying method, and recommended brining process and mechanical drying method. Three types of fish commonly used for fish drying were bought in Roxas City Fish Port, such as Sardine (*Sardinella Spp*), Anchovies (*Stolephorus comersonii*) and Red Mullet (*Mullus surmuletus*). Each of the fish types were prepared in three treatments such as T1 – traditional brining + sun-drying, T2 – DOST/ITDI recommended brining + CAPSU Fish Dryer, and T3 - traditional brining + CAPSU Fish Dryer. The population of APC recorded in this study varies from 1,200,000 to 1,900,000 cfu/g for samples taken from traditionally brined and sun-dried fish. Samples taken from mechanically dried fish, both brined using the traditional and DOST-ITDI recommended methods, resulted to APC of 380 to 6,200 cfu/g. These values are within the maximum count achievable under GMP. All the tested samples from mechanical drying brined using the traditional and DOST-ITDI recommended methods showed a coliform count less than 10 MPN/g, hence passed the GMP standards. Yeast and mold count were recorded to be less than <100 cfu/g in mechanically dried and DOST-ITDI brined Sardine and Anchovies while for Split Red Mullet, yeast and mold count is 200 and 230 cfu/g for traditionally brined and DOST-ITDI based brining, respectively. Most samples tested in the study passed the minimum standard value of 12 fo NaCl as Salt content. All the tested samples showed a water activity value below 0.78. This shows that mechanical drying can be used to lower water activity levels of fish. Based on the level of bacterial load, all sun-dried samples were heavily loaded with bacteria irrespective of the fish type and the brining methods. A pronounced load of coliforms was observed in Anchovies samples which could be sourced from contaminated water. The recorded water activity level of all samples <0.78 could reflect that the drying process, both sun-drying and mechanical drying is sufficient to remove the free water which could have been used by pathogenic bacteria growth and toxin production. It is generally noted that although dried fish could serve as a nutritious component in a diet, the microbial quality of products in traditional processing and drying is improved by mechanical drying.

Keywords: Dried Fish, Microorganisms, Water Activity, Contamination

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

Fish is an important foodstuff due to its nutritional value as a main source of protein. In the Philippines, the mean per capita consumption of fish and fishery products in 2017 was 40

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 3 of 10

kg/year or 109 grams/day, representing 12.8% of the total food intake. Based on the SEAFDEC report by Lamarca N. (2017), around 70% of the country's fish catch is consumed fresh or chilled, while 30% is processed. Fish is a highly perishable food that invariably becomes rotten within a few hours after capture. Hence, the need for preservation and processing to reduce microbial decomposition and retard spoilage.

Various methods have been used to ensure the safety and prolong the shelf life of fish while maintaining quality. Although cold storage is the preferred technique, the limitations of standard storage facilities like refrigerators often drive small-scale fish producers to seek alternative preservation technologies. Fermentation, drying, and smoking are some of the most commonly used methods worldwide for long-term fish preservation. "In the Philippines, fish drying is the most practiced processing method, especially for the small-sized species". It was found that reducing the moisture content of fish to 25% will retard bacteria and reduce autolytic activity.

Sun drying is the widely used method for fish drying as it requires cheaper operating expenses. However, this method has disadvantages due to unpredictable weather conditions, resulting in uneven drying and a relatively slow drying rate, thus the tendency of spoilage. In addition, traditional drying methods are often associated with poor hygienic conditions, often resulting in microbial contamination. Besides hygiene issues, factors such as exposure to air, dust, contaminated water, and soil can also negatively impact the quality of dried fish. The presence of a high number of microorganisms, particularly harmful pathogens, can pose risks to consumer health.

Hence, several dryers have been developed to dry fish products. Biomass-fired dryer, for example, is an effective alternative to traditional sun-drying and dryers, which use expensive electricity and fossil fuels for their heat source. Biomass is a clean and environmentally friendly fuel and does not cause any pollution during combustion. It has been used successfully as a fuel in many agricultural machines – including a flatbed dryer for grains. It entails cheap investment and operating costs. Rice husk, a by-product of rice milling, is an attractive biomass fuel because it is readily available in large quantities and easy to collect. Hence, the CAPSU Biomass-Fired Fish dryer has been developed. In addition to faster drying rates, one advantage of this technology is ensuring hygienic drying operations.

Aside from drying methods, the brining process also plays a crucial role in determining the quality of dried fish. Brining involves soaking the fish in a saltwater solution, which is intended to enhance preservation and flavor. However, the effectiveness of this process and the resulting quality of the dried fish are influenced by several factors. The concentration of salt in the brine solution affects how well the fish is preserved. A higher salt concentration can improve preservation and flavor but may result in overly salty dried fish. Conversely, a lower salt concentration might not provide sufficient preservation, leading to potential spoilage and decreased quality. Likewise, the length of time the fish spends in the brine solution is also important. Adequate brining ensures that salt is evenly distributed throughout the fish, which helps in effective preservation. Insufficient brining can lead to uneven salting, which can

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 4 of 10

impact the texture, flavor, and safety of the dried fish. Proper brining is essential for ensuring optimal preservation and quality in the final dried product.

Traditional fish sun-drying often involves the use of traditional brining methods as part of the preservation process. In this approach, fish are first soaked in a saltwater solution to enhance their preservation and flavor before being dried in the sun. On the other hand, the Department of Science and Technology has brining recommendations for different dried fish types.

In summary, both the brining process and the drying methods contribute to the overall quality of dried fish.

Evaluating the level of contamination is crucial in determining the safety of dried fish and ensure it is safe for consumption. Assessments of microbial loads are commonly employed to gauge the safety of dried fish products. Additionally, the water activity is also a helpful indicator of the shelf life of products.

There is limited data on the microbial safety of dried fish products processed by sun-drying and mechanical biomass-fired dryer. Hence, this study aimed to assess the water activity, salt content and microbial load of locally prepared dried fish using traditional brining and sun-drying method, and recommended brining process and mechanical drying method.

Lastly, this project will improve scientific dissemination and serve as a direct source of knowledge or ideas related to dried fish processing towards a more advanced technological possibilities.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

a. Drying preparations

Three types of fish commonly used for fish drying were bought in Roxas City Fish Port, such as Sardine (*Sardinella Spp*), Anchovies "Dilis" (*Stolephorus comersonii*) and Red Mullet (*Mullus surmuletus*). Each of the fish types were prepared in three treatments such as:

T1 – traditional brining + sun-drying,
T2 – DOST/ITDI recommended brining + CAPSU Fish Dryer, and
T3 - traditional brining + CAPSU Fish Dryer.

Twenty kilograms of each fish types were prepared for each treatment.

The traditional brining for Sardine (*Sardinella Spp*) was done by applying salt into the fish at 1:3 (fish:salt) ratio. It was left for twelve hours to allow absorption. The fish were then washed by seawater done by dipping the fish using a strainer. Dipping was done twice or thrice until salt are removed from fish. The same method was used for Mullet

	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 5 of 10

(*Mullus surmuletus*) after the fish were splitted and eviscerated. For Anchovies "Dilis" (*Stolephorus comersonii*), brining was done by dipping fish into a bucket of seawater twice.

The DOST-ITDI recommended brining for Sardine (*Sardinella Spp*) was done by soaking the fish into 25% brine solution for 3 hours. The fish were then drained and rinsed with clean fresh water to remove surface salt. For split Mullet (*Mullus surmuletus*), brining was done by soaking the fish into 20% brine solution for 1 hour. The fish were then drained and rinsed with clean fresh water to remove surface salt. On the other hand, the brining of Anchovies (*Stolephorus comersonii*) was done by dipping/rinsing fish into 20% brine solution in a bucket twice.

b. Fish drying

For sun-drying, fish were laid into bamboo-slats made fish trays, while for CAPSU biomass dryer, the fish were laid in a stainless steel fish trays. Sardine (*Sardinella Spp*) and Anchovies (*Stolephorus comersonii*) were laid flat on trays, while spilt Mullet (*Mullus surmuletus*) were laid in the trays with the skin down. Right spacing was done in laying the fish into the trays wherein heads and tails were laid alternately to maximize space.

For sun-drying, the trays with fish were laid into elevated bamboo stands under the sun until moisture content is reduced to suitable level. The fish were turned every five hours for uniform drying. For CAPSU biomass dryer, the dryer was pre-heated to desired air temperature, and the fish were loaded into the drying chamber until moisture content is reduced to suitable level. No turning of fish is required for mechanical drying.

c. Sample Collection

Dried fish samples weiging two hundred fifty grams (250g) were randomly collected after drying and post-drying processes. Each sample were packed, labelled, and sealed. The prepared samples were then submitted to the DOST Region VI – Resgional Standards and Testing Laboratory for the following tests:

1. Water Acticity
2. NaCl as Salt
3. Aerobic Plate Count
4. Total Coliform Count
5. Molds and Yeast

d. Water Activity was determined by Novasina Water Activity Meter based on TMP-CHEM-023. Analysis was done in triplicates.

e. Salt as Sodium Chloride was determined by Volumetric Method, an Official Method of Analysis of AOAC INTERNATIONAL (2019) 21st Ed., Official Method 937.09. Analysis was done in triplicates.

f. Microbial Analysis

	Document Type:	DOCUMENTED INFORMATION		Document Code	RES- CR-01
		ISO 9001:2015		Revision No.	00
	Document Title:	COMPLETED RESEARCH		Effective Date	
				Page	Page 6 of 10

Aerobic Plate Count was determined using the Pour Plate Method at 35°C for 48 hrs based on PCA, USFDA BAM Online (2001).

Coliform Count was determined using the Multiple Tube Fermentation Technique based on USFDA BAM Online (2017).

Mold and Yeast Count was determined using the Spread Plate Method, 25°C for 5-7 days based on DRBCA, USFDA BAM Online (2001).

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

A. Aerobic Plate Count (APC)

The aerobic plate count is one of the indicators for the quality of dried fish. The population of APC recorded in this study varies from 1,200,000 to 1,900,000 cfu/g for samples taken from traditionally brined and sun-dried fish. Similar studies were conducted in dried fish and reported total bacterial count ranging from 10^5 to 10^9 [1]. The total bacterial load generally indicates the poor microbiological quality of all the tested samples. The bacterial load of all tested samples under this treatment are below the standard [2]. On the other hand, samples taken from mechanically dried fish, both brined using the traditional and DOST-ITDI recommended methods, resulted to APC of 380 to 6,200 cfu/g. These values are within the maximum count achievable under GMP.

Table 1. Aerobic Plate Count (APC).

Type of Dried Fish	Brining Method	Drying Method		Standard Value (cfu/g)
		<i>Mechanical Drying</i>	<i>Sun-drying</i>	
Sardine "Tabagak" (<i>Sardinella Spp</i>)	a. Traditional Brining	380 cfu/g	1,200,000 cfu/g	m = 100,000 M = 500,000
	b. 25% brine concentration, soaked for 2-3 hrs*	<250 cfu/g		
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	a. Traditional Brining	1,000 cfu/g	1,900,000 cfu/g	m = 100,000 M = 500,000
	b. 25% brine concentration, soaked for 30 seconds*	300 cfu/g		
Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	a. Traditional Brining	6,200 cfu/g	1,200,000 cfu/g	m = 100,000 M = 500,000
	b. 20% brine concentration, soaked for 1 hrs*	5,200 cfu/g		

*DOST-ITDI Recommended Brining Method

m is the maximum count achievable under GMP; and

M is the maximum count beyond which product safety /quality may be affected

B. Total Coliform Count

	Document Type:	DOCUMENTED INFORMATION		Document Code	RES- CR-01
		ISO 9001:2015		Revision No.	00
	Document Title:	COMPLETED RESEARCH		Effective Date	
				Page	Page 7 of 10

In this study, coliforms were detected in all the tested dried fish samples. All the tested samples from mechanical drying brined using the traditional and DOST-ITDI recommended methods showed a coliform count less than 10 MPN/g, hence passed the GMP standards. On the other hand, the Anchovies (*Stolephorus comersonii*) samples had a total coliform ranging of 1,700 MPN/g. Although similar finding has been reported from previous research [3], standards declare that coliforms should not exceed 100 in a quality dried fish product [4]. The water could be the major source for the contamination by coliforms. Anchovies (*Stolephorus comersonii*) were brined using seawater based on traditional processing. Besides, lack of hygiene during the drying process which might include the usage of non-standard processing equipment and poor handling of the fish might have contributed to the presence of coliforms in such numbers.

Table 2. Total Coliform Count (TCC) based on Analysis Report.

Type of Dried Fish	Brining Method	Drying Method		Standard Value (MPN/g)
		Mechanical Drying	Sun-drying	
Sardine "Tabagak" (<i>Sardinella Spp</i>)	a. Traditional Brining	<1.8 MPN/g	<1.8 MPN/g	m = 10
	b. 25% brine concentration, soaked for 2-3 hrs*	<1.8 MPN/g		M = 100
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	a. Traditional Brining	7.8 MPN/g	1,700 MPN/g	m = 10
	b. 25% brine concentration, soaked for 30 seconds*	<1.8 MPN/g		M = 100
Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	a. Traditional Brining	<1.8 MPN/g	<1.8 MPN/g	m = 10
	b. 20% brine concentration, soaked for 1 hr*	<1.8 MPN/g		M = 100

*DOST-ITDI Recommended Brining Method

m is the maximum count achievable under GMP; and

M is the maximum count beyond which product safety /quality may be affected

C. Mold and Yeast

Yeast and mold count were recorded to be less than <100 cfu/g in mechanically dried and DOST-ITDI brined Sardine and Anchovies while for Split Red Mullet, yeast and mold count is 200 and 230 cfu/g for traditionally brined and DOST-ITDI based brining, respectively. Higher than GMP standard yeast and mold count is recorded from Anchovies and Split Red Mullet samples which were sun-dried. The presence of yeasts and molds in such dried food products is because of their higher tolerance in reduced water activity levels in contrast to pathogenic bacteria which does not grow at water activity levels below 0.86 [5].

Table 3. Mold and Yeast Count (MYC) based on Analysis Report.

Type of Dried Fish	Brining Method	Drying Method
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	Document Type:	DOCUMENTED INFORMATION		Document Code	RES- CR-01
		ISO 9001:2015		Revision No.	00
	Document Title:	COMPLETED RESEARCH		Effective Date	
				Page	Page 8 of 10

		Mechanical Drying	Sun-drying	Standard Value (cfu/g)
Sardine "Tabagak" (<i>Sardinella Spp</i>)	a. Traditional Brining	<100 cfu/g	<100 cfu/g	m = 1,000
	b. 25% brine concentration, soaked for 2-3 hrs*	<100 cfu/g		M = 10,000
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	a. Traditional Brining	100 cfu/g	2,600 cfu/g	m = 1,000
	b. 25% brine concentration, soaked for 30 seconds*	<100 cfu/g		M = 10,000
Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	a. Traditional Brining	200 cfu/g	4,400 cfu/g	m = 1,000
	b. 20% brine concentration, soaked for 1 hrs*	230 cfu/g		M = 10,000

*DOST-ITDI Recommended Brining Method

m is the maximum count achievable under GMP; and

M is the maximum count beyond which product safety /quality may be affected

D. NaCl as Salt

Most samples tested in the study passed the minimum standard value of 12 fo NaCl as Salt content. On the other hand, using the traditional brining method for Anchovies and the DOST-ITDI recommended brining method for Sardines did not pass the standard requirement for NaCl content.

Table 6. NaCl as Salt Value based on Analysis Report.

Type of Dried Fish	Brining Method	Drying Method		Standard Value
		Mechanical Drying	Sun-drying	
Sardine "Tabagak" (<i>Sardinella Spp</i>)	Traditional Brining	12.98	15.44	minimum of 12
	25% brine concentration, soaked for 2-3 hrs*	7.03		
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Traditional Brining	6.16	7.12	minimum of 12
	25% brine concentration, soaked 30 seconds*	23.26		
Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	Traditional Brining	15.93	13.42	minimum of 12
	20% brine concentration, soaked for 1 hrs*	14.32		

*DOST-ITDI Recommended Brining Method

E. Water Activity

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 9 of 10

All the tested samples showed a water activity value below 0.78. This shows that mechanical drying can be used to lower water activity levels of fish. Water activity is an important factor in food preservation since it provides information on the quality deterioration including the growth of microorganisms [5, 6]. Most pathogenic bacterial growth and toxin production are highly hindered at low water activity.

Table 7. Water Activity based on Analysis Report.

Type of Dried Fish	Brining Method	Drying Method		Standard Value
		Mechanical Drying	Sun-drying	
Sardine "Tabagak" (<i>Sardinella Spp</i>)	Traditional Brining	0.7	0.71	maximum of 0.78
	25% brine concentration, soaked for 2-3 hrs*	0.74		
Anchovies "Dilis" (<i>Stolephorus comersonii</i>)	Traditional Brining	0.65	0.65	maximum of 0.78
	25% brine concentration, soaked for 2-3 hrs*	0.64		
Split Red Mullet "Daing Salmonete" (<i>Mullus surmuletus</i>)	Traditional Brining	0.7	0.76	maximum of 0.78
	20% brine concentration, soaked for 1 hrs*	0.71		

*DOST-ITDI Recommended Brining Method

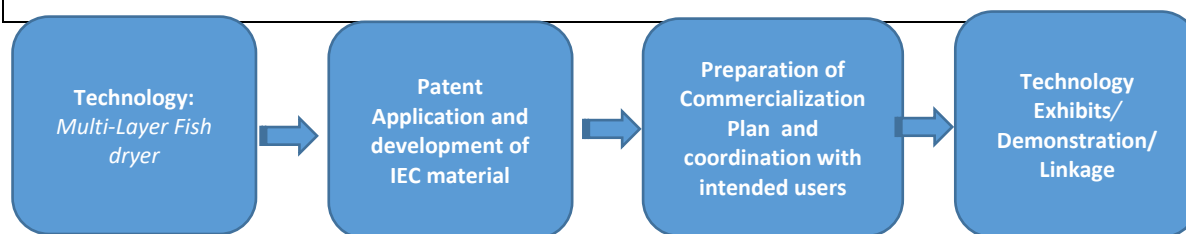
(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

The microbiological quality of traditional and DOST-ITDI based brining, and sun-dried and mechanical dried fish were studied. Based on the level of bacterial load, all sun-dried samples were heavily loaded with bacteria irrespective of the fish type and the brining methods. A pronounced load of coliforms was observed in Anchovies samples which could be sourced from contaminated water, unsanitary utensils and/or poor hygienic handling during the preparation of the dried fish. The recorded water activity level of all samples <0.78 could reflect that the drying process, both sun-drying and mechanical drying is sufficient to remove the free water which could have been used by pathogenic bacteria growth and toxin production. It is generally noted that although dried fish could serve as a nutritious component in a diet, the microbial quality of products in traditional processing and drying is improved by mechanical drying.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

- 1 Prepared paper for publication
- 1 IEC material
- 2 People Services (Conducted training on the use of the dryer among intended users)

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)



	Document Type:	Document Code	RES- CR-01
	DOCUMENTED INFORMATION ISO 9001:2015	Revision No.	00
	Document Title:	Effective Date	
	COMPLETED RESEARCH	Page	Page 10 of 10

(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.)

N/A

(17) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

Gratitude is extended to the DOST-TAPI, for funding the study, and to CapSU the facilities and other resources.

(18) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

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[6] Justine Maiworé, Leopold Tatsadjieu Ngoune, Mahi Koné Koumba, Isabelle Metayer, Didier Montet, Noël Durand, Determination of bacterial population and the presence of pesticide residues from some Cameroonian smoked and dried fish, *Scientific African*, Volume 13, 2021, e00886, ISSN 2468-2276, <https://doi.org/10.1016/j.sciaf.2021.e00886>.

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